higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

ELECTROTECHNICS N6

9 APRIL 2019

This marking guideline consists of 10 pages.

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QUESTION 1: DC MACHINES

1.1
- Variation of the flux
- Armature or Rheostatic control method
- Series-parallel control
- Voltage control (4)

1.2 Given: A d.c. shunt motor

$V = 250\ V$ $N_1 = 1\ 500$ r/min

$R_f = 250\ \Omega$ $I_a = 20\ A$

$R_a = 0,25\ \Omega$.

$R_{ser} = 250\ \Omega$

$$I_{f1} = \frac{V}{R_f} = \frac{250}{250} = 1\ A$$

$$E_1 = V - I_a R_a = 250 - (20 \times 0,25) = 250 - 5 = 245\ V$$

1.2.1
$$I_{f2} = \frac{V}{R_f + R_{ser}} = \frac{250}{(250+250)} = 0,5\ A$$

$T_2 = T_1$ (load torque remaining the same)

$$I_{a2}\varnothing_2 = I_{a1}\varnothing_1$$

$$\therefore\ I_{a2} = \frac{I_{a1}\varnothing_1}{\varnothing_2} = I_{a1} \times \frac{I_{fi}}{I_{f2}} = 20 \times \frac{1}{0,5} = 40\ A \qquad (\varnothing \propto I_f)$$

(5)

1.2.2
$$E_2 = V - I_{a2} R_a = 250 - (40 \times 0,25) = 250 - 10 = 240\ V$$

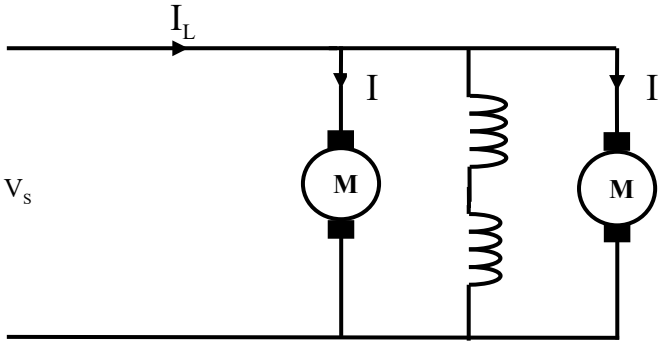
$$\frac{N_2\varnothing_2}{N_1\varnothing_1} = \frac{E_2}{E_1}$$

$$\therefore\ N_2 = \frac{E_2}{E_1} \times \frac{N_1\varnothing_1}{\varnothing_2} = \frac{240}{245} \times \frac{1\ 500 \times 1}{0,5}$$

$$= 2\,938,776\ r/min$$

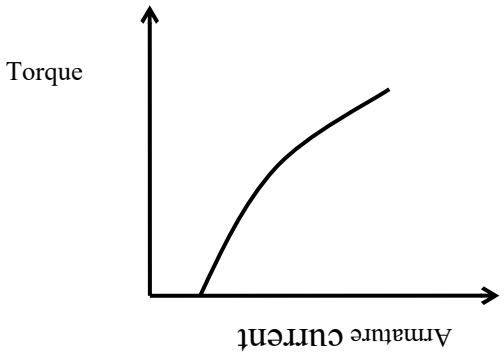
(3)

1.3



(2)

1.4



(1)
[15]

QUESTION 2: AC CIRCUIT THEORY

Given: A balanced 3-phase star-connected load

Z = (8 + j 6) ohms/phase

V_L = 230 V supply.

2.1 $Z=\sqrt{R^2+X^2}=\sqrt{8^2+6^2}$

$=10\ \Omega$ ✓

$I_L=I_{PH}=\frac{V_{PH}}{Z}=\frac{132,791}{10}=13,279\ A$

$V\ ph=\frac{V_L}{\sqrt{3}}=\frac{230}{\sqrt{3}}=132,791\ V$

(5)

2.2 $pf=\frac{R}{Z}=\frac{8}{10}=0,8\ Lagging$

(2)

2.3 $P_T=\sqrt{3}\cdot V_L\cdot I_L\cos\theta$

¿$\sqrt{3}\times 230\times 13,279\times 0,8$

$=4\,231,982\ W$✓✓

$P_T=3\cdot V_{PH}\cdot I_{PH}\cos\theta$

$=3\times 132,791\times 13,279\times 0,8$

$=4\,231,996\ W$✓✓

(2)

2.4 $Q=\sqrt{3}\cdot V_L\cdot I_L\cdot\times\sin\theta$

$(\sqrt{3})\ (230)(13,279)(\sin\ 36,87°)$

¿$3\,174\ VA_r$✓✓

$Q=3\cdot V_{PH}\cdot I_{PH}\cdot\sin\theta$

$=(3)(132{,}791)(13{,}279)(\sin 36,87°)$

¿$3\ 174\ VA_r$✓✓

(2)

2.5 $S=\sqrt{P^2+Q^2}=\sqrt{(4\,231{,}982)^2+(3\,174)^2}=5$

$S=3\cdot V_{PH}\cdot I_{PH}$

$=3\times 132,791\times 13.279$

$=5\,289,996\ VA$✓

(2)
[13]

QUESTION 3: TRANSFORMERS

3.1
- Increase in core loss
- Increase in copper loss due circulating currents
- Magnetic interference with communication circuits and protective devices

(Any 2 × 1) (2)

3.2 Voltage regulation of a transformer is the rise or fall in the secondary✓ voltage when the load is removed and the primary voltage remains the same.✓

OR

Voltage regulation of a transformer is the difference in the secondary terminal voltage✓ between no-load and full-load,✓ expressed as a ratio of the no-load voltage with the primary voltage remaining constant. (2)

3.3 **Given:** a single-phase transformer

S = 25 kVA V_2 = 250 V

V_1 = 2 500 V % Z = (1,2 + j3,5)%

pf = 0,8 lagging

3.3.1

Method 1

$S = V_1 \cdot I_1$

$\therefore\ I_1 = \frac{S}{V_1} = \frac{25 \times 10^3}{2\ 500} = 10\ A$

$\%R = \frac{I_1 \cdot R_{e1}}{V_1}$

$\therefore\ R_{e1} = \frac{\%R \cdot V_1}{I_1} = \frac{0{,}012 \times 2\,500}{10}$

$= 3\ \Omega$

$P_{sc} = I_1^2 \cdot R_{e1} = 10^2 \times 3 = 300\ W$

$\frac{S \cdot \cos\theta}{S \cdot \cos\theta + P_o + P_{sc}} \times 100\%$

$\frac{25 \times 0{,}8}{(25 \times 0{,}8) + 0{,}23 + 0{,}3} \times 100\%$

$= 97{,}418\ \%$

Method 2

$S = V_1 \cdot I_2$

$\therefore\ I_2 = \frac{S}{V_2} = \frac{25 \times 10^3}{250} = 100\ A$

$\%R = \frac{I_2 \cdot R_{e2}}{V_2}$

$\therefore\ R_{e2} = \frac{\%R \cdot V_2}{I_2} = \frac{0{,}012 \times 250}{100}$

$= 0{,}03\ \Omega$

$P_{sc} = I_2^2 \cdot R_{e2} = 100^2 \times 0{,}03$

$= 300\ W$

(4)

3.3.2 $P_o = k^2 \cdot P_{sc}$ (2)

$$k=\sqrt{\frac{P_o}{P_{sc}}}=\sqrt{\frac{0,\ 23}{0,\ 3}}=0,\ 876$$

Method 1

$$¿\frac{kS\cdot\cos\theta}{kS\cdot\cos\theta+P_o+k^2P_{sc}}\times100\%$$

$$¿\frac{0,\ 876\times25\times0,\ 8}{(0,\ 876\times25\times0,\ 8)+0,\ 23+(0,\ 876^2\times0,\ 3)}\times100\%$$

$$=97,\ 441\ \%$$

Method 2

$$¿\frac{kS\cdot\cos\theta}{kS\cdot\cos\theta+2P_o}\times100\%$$

$$¿\frac{0,\ 876\times25\times0,\ 8}{(0,\ 876\times25\times0,\ 8)+(2\times0,\ 23)}\times100\%$$

$$¿97,\ 442\ \%$$

3.3.3

$$\%X=\frac{I_1\cdot X_{e1}}{V_1}$$

$$\therefore\ X_{e1}=\frac{\%X\cdot V_1}{I_1}=\frac{0{,}035\times2500}{10}$$

$$=8,\ 75\ \Omega$$

Method 1

$$Z_{e1}=R_{e1}+X_{e1}=3+j8,\ 75$$

$$=9,\ 25\ \Omega$$

$$V_{1sc}=I_1\cdot Z_{e1}=10\times9,\ 25\ =92,\ 5\ V$$

Method 2

$$Z_{e1}=\sqrt{{R_{e1}}^2+{X_{e1}}^2}=\sqrt{3^2+8,\ 75^2}$$

$$¿9,\ 25\ \Omega$$

(3)

3.3.4 **Method 1**

$$Voltage\ \ Regulation=\frac{I_1(R_{e1}\cdot\cos\theta+X_{e1}\cdot\sin\theta}{V_1}\times100\%$$

$$¿\frac{10[(3\times0,\ 8)+(8,\ 75\times0,\ 6)}{2500}\times100\%$$

$$=3,\ 06\ \%$$

Method 2

$$Voltage\ \ Regulation\ =\%R\cos\theta+\%X\cdot\sin\theta$$

$$¿(1,\ 2\%\times0,\ 8)+(3{,}5\%\times0,\ 6)$$

$$=3,\ 06\%$$

(2)
[15]

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QUESTION 4: AC MACHINES - ALTERNATORS

4.1 **Given:** A 1 500 kVA, 6,6 kV, three-phase, Y-connected alternator

R = 0,4 ohms

X = 6 ohms per phase

p.f = 0,8 lagging

$$I=\frac{P}{\sqrt{3}V_L}=\frac{1\ 500\times10^3}{\sqrt{3}\times6\ 600}=131,\ 216\ A$$

$$IR=131,\ 216\times0,\ 4=52,\ 486\ V$$

$$IX=131,\ 216\times6=787,\ 296\ V$$

$$V_{PH}=\frac{V_L}{\sqrt{3}}=\frac{6\ 600}{\sqrt{3}}=3810,\ 512\ V$$

$$\varnothing=\cos^{-1}(0,\ 8)=36,\ 87°$$

$$\tan(\varnothing+\alpha)=\frac{AB}{OA}=\frac{V\sin\varnothing+IX}{V\cos\varnothing+IR}$$

$$¿\frac{(3810,\ 512\times0,\ 6)+787,\ 296}{(3810,\ 512\times0,\ 8)+52,\ 486}$$

$$=0,\ 991$$

$$(\varnothing+\alpha)=\tan^{-1}(0,\ 991)$$

$$=44,\ 741°$$

$$\therefore\ \alpha=44,\ 741°-36,\ 87°$$

$$=7,\ 871°$$

(9)

4.2 Armature reaction is the influence of the stator ampere-turns on the value and distribution of the magnetic flux✓ in the air-gaps between the poles and stator core.✓ (2)

4.3 Synchronous impedance is when the synchronous reactance✓ and armature resistance✓ of an alternator are vectorially combined. (2)

[13]

QUESTION 5: AC MACHINES - SYNCHRONOUS MOTORS

5.1
- Hunting is a condition produced when a synchronous motor is used for driving varying loads.
- To minimize the buildup of these oscillation dampers or damping grids are used.
- These dampers consists of short-circuited copper bars embedded in the faces of the field poles of the motor.
- The oscillatory motion of the rotor sets up eddy currents in the dampers which flow in such a way as to suppress these oscillations. (4)

5.2 **Given** A single-phase synchronous motor.

V_L = 380 V P_{motor} = 36,5 kW

R_a = 0,2 Ω pf = 0,8 leading

X_s = 3 Ω efficiency = 80%

5.2.1

$$\eta \frac{P_{out}}{P_{in}}$$

$$\therefore \quad P_{in} = \frac{P_{out}}{\square} = \frac{36,5}{0,8} = 45,625 \ kW$$

$$P_{in} = \sqrt{3} \cdot V_L \cdot I_L \cdot \cos\theta$$

$$\therefore \quad I_L = \frac{P_{in}}{\sqrt{3} \cdot V_L \cdot \cos\theta} = \frac{45,625 \times 10^3}{\sqrt{3} \times 380 \times 0,8} = 86,65 \ A$$

(3)

5.2.2

$$V_P = \frac{V_L}{\sqrt{3}} = \frac{380}{\sqrt{3}} = 219,393 \ V$$

$$Z_s = R_a + X_s$$

$$= 0,2 + j3$$

$$= 3,00686,186° \ \Omega$$

$$\cos\theta = 0,8$$

$$\therefore \ \theta = \cos^{-1} 0,8 = 36,87°$$

$$E_r = I Z_S$$

$$= 86,65 \ 36,87 \times 3,00686,186$$

$$= 260,47123,056° \ V$$

$$E_p = E_R + V_P$$

$$= 260,47123,056 - 219,3930°$$

$$= 422,278148,87° \ V$$

$$E_L = \sqrt{3} \cdot E_p = \sqrt{3} \times 422,278 = 731,407 \ V$$

Method 2

$I R_a = 86,65 \times 0,2 = 17,33\ V$

$I X_s = 86,65 \times 3 = 259,95\ V$

$E_p = V_p \text{-} \theta + I R_a 180° + I X_s \text{-} 90°$
$= 219,393 \text{-} 36,87° + 17,33 180° + 259,95 \text{-} 90°$
$= 422,329 \text{-} 68°\ V$

- $E_L = \sqrt{3} \cdot E_p = \sqrt{3} \times 422,329 = 731,495\ V$

(6)

5.3
- A 'pony motor'
- Damping grids

(2)
[15]

QUESTION 6: AC MACHINES - INDUCTION MOTORS

6.1 **Given:** A three-phase induction motor
Number of poles = 4-pole
f = 50 Hz

6.1.1 $N_S = \frac{60 \times f}{p} = \frac{60 \times 50}{2} \checkmark = 1500\ r/min \checkmark$

(2)

6.1.2 $\%S = \frac{N_s - N_r}{N_s}$

$0,04 = \frac{1500 \text{-} N_r}{1500}$

$\therefore N_r = 1500 \text{-} (0,04 \times 1500)$
$= 1440\ r/min$

(3)

6.1.3 $p.uS = \frac{1500 \text{-} 600}{1500} = 0,6 \checkmark$

$\therefore rotor\ \ frequency = Sf = 0,6 \times 50 = 30\ Hz$

(3)

6.2 Synchronous watt - is that torque✓ which, at the synchronous speed of the machine under consideration, would develop a power of one watt.✓ (2)

6.3
- direct starting
- star-delta starting
- auto-transformer starting
- slip-ring motor starting

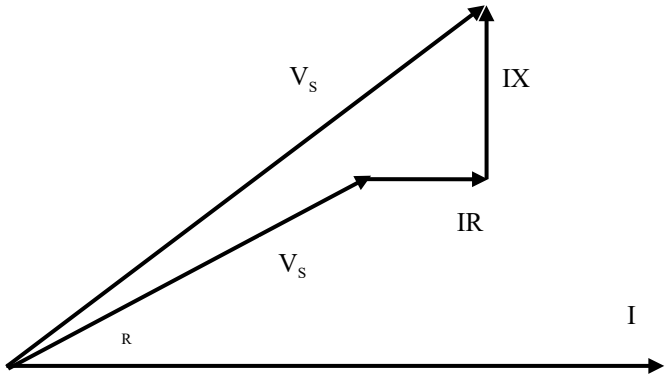
(4)
[14]

QUESTION 7: GENERATION AND DISTRIBUTION OF AC

7.1
- it decreases the efficiency of both apparatus and supply system
- It limits the output of both generators and transformers
- It causes a greater fall in terminal voltage (Any 2 × 1) (2)

7.2
- The consumer pays a lower tariff
- The efficiency of their installation is high (2)

7.3 **Given:** A 3-phase transmission line



V_R = 11 kV
R = 5 Ω/phase
L = 30 mH/phase
pf = 0,8 lagging
P_{out} =1 000 kW
f = 50 Hz

7.3.1 $$V_R = \frac{V_L}{\sqrt{3}} = \frac{11\ 000}{\sqrt{3}} = 6\,350,\ 853\ V$$

$$I_L = \frac{P}{\sqrt{3}\ V_L I_L \cos\varnothing} = \frac{1\ 000 \times 10^3}{\sqrt{3} \times 11\ 000 \times 0,\ 8} = 65,\ 608\ A$$

$$X_L = 2\pi fL = 2\pi \times 50 \times 30 \times 10^{-3} = 9,\ 425\ \Omega \checkmark$$

Z_{ph} = (5 + *j* 9.425) Ω

$V_{drop} = IZ_{ph}$ = 65,608 (5 + j 9,425)

= 699, 981∠62, 054° V ✓

OR

IR = 65, 608 x 5 = 328,04 V

IX = 65, 608 x 9,425 = 618,355 V

V_{drop} = IR + IX = 328,04 + j 618,355

= 700,028∠62,047° V

$V_{S/ph} = V_R + V_{drop}$ = 6 350,853 + (328,04 + j 618,355)

= 6 678,893 + j 618,355

= 6 707,457∠5,29° V✓

∴ $V_{S/Line} = \sqrt{3} \times V_{S/ph} = \sqrt{3}$ x 6 707,457

= 11 617,656 V ✓ (6)

7.3.2 $\cos\theta_R = 0,\ 8 \quad \therefore \quad \theta_R = 36,\ 87°$

$\varnothing_S = 36,\ 87° + 5,\ 29° = 42,\ 16° \quad \therefore \quad \cos\varnothing_S = 0,\ 741$ Lagging ✓ (1)

7.3.3 $P_{Total\ Loss} = 3I^2R = 3 \times 65,\ 608^2 \times 5 = 64,\ 566{,}145\ kW$ √

$P_{input} = P_{output} + P_{Loss} = 1\,000 + 64{,}566 = 1\,064,\ 566\ kW$ √

$\therefore\ \eta_{power\ transmission} = \dfrac{1\ 000}{1\ 064,\ 566} \times 100\% = 93,\ 935\%$ (2)

7.3.4 $\therefore\ \%\ regn. = \dfrac{V_S - V_R}{V_R} = \dfrac{11\,617,\ 656 - 11\,000}{11\,000} \times \dfrac{100\%}{1}$

$= 5,\ 615\%$ (2)

[15]

TOTAL: 100